Session 21 — Leases and Deferred Taxes: Specialized Cases (Sale and Leaseback, Tax Rate Changes)

By the end of this session you must know what a sale-and-leaseback transaction is and why both parties enter into one, how to account for it on both the seller-lessee's and buyer-lessor's side depending on whether the transfer qualifies as a sale, and how a change in the statutory tax rate is reflected in existing deferred tax asset/liability balances.

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1. Sale and Leaseback: What and Why?

Sale and leaseback: the seller-lessee sells an asset to the buyer-lessor and then immediately leases that same asset back from the buyer-lessor.

Must know (why each side does this):

• **Seller-lessee's benefits:** an immediate cash inflow that can be redeployed elsewhere in the business, with no interruption to operations (the seller-lessee never loses use of the asset)

• **Buyer-lessor's benefits:** a steady, relatively low-risk income stream from a known counterparty whose credit risk can be investigated directly; an attractive yield, typically above market averages, because the seller-lessee places high value on retaining operational control of the asset; and potentially important tax advantages

2. Accounting Treatment: Seller-Lessee Perspective

- The accounting treatment turns entirely on one question: does the transfer of the asset qualify as a sale under IFRS15 (i.e. has control of the asset actually transferred)?

Must know (seller-lessee, by transfer type):

• **If the transfer IS a sale** (control transfers): the seller-lessee derecognises the transferred asset and recognises a right-of-use asset plus a corresponding lease liability. The ROU asset is measured initially at the *proportion* of the asset's previous carrying amount that relates to the rights-of-use *retained* by the seller-lessee. The profit or loss on disposal is restricted in proportion to the rights actually *transferred* to the buyer-lessor — i.e. the seller-lessee only recognises a gain/loss on the portion of the asset it has genuinely given up, not on the whole asset

• **If the transfer is NOT a sale** (control does not transfer): the seller-lessee continues to recognise the transferred asset unchanged, and the "sale" proceeds received are instead treated as a loan, recognised as a financial liability under IFRS9

3. Accounting Treatment: Buyer-Lessor Perspective

Must know (buyer-lessor, by transfer type):

• **If the transfer IS a sale:** the buyer-lessor accounts for the purchase of the asset under the relevant standard (e.g. IAS16 for PP&E), and separately accounts for the lease it grants back to the seller-lessee under IFRS16

• **If the transfer is NOT a sale:** the buyer-lessor does not recognise the asset at all; the proceeds it paid are instead treated as a loan, recognised as a financial asset under IFRS9

- Notice the symmetry: whichever side of the sale/not-a-sale line the transaction falls on, both parties treat it consistently — either both apply asset derecognition/recognition plus lease accounting, or both treat the cash transfer as a loan (financial liability for the seller-lessee, financial asset for the buyer-lessor) under IFRS9.

4. Worked Example: Sale and Leaseback

Setup: on 1 July, a seller-lessee sells a building to a buyer-lessor for €500,000 (its fair value); immediately before the sale, the building's carrying amount was €400,000. On the same date, the seller-lessee leases the building back for 5 years, with lease payments (at market rate) having a present value of €105,000. The transfer qualifies as a sale under IFRS15.

Watch out: the extracted slides pose this example ("explain how this transaction should be accounted for") without a printed numerical solution. The walkthrough below applies the method from Section 2 directly, and the resulting balance sheet effect is internally consistent (assets change = liabilities change + gain recognised), which is a strong self-check, but it is a self-constructed application, not a transcription of an official answer key.

Applying the method: the proportion of rights *retained* by the seller-lessee is approximated by the ratio of the leaseback's present value to the asset's fair value: 105,000 ÷ 500,000 = 21%.

ROU asset = previous carrying amount × proportion retained = 400,000 × 21% = **€84,000**.
Lease liability = present value of the lease payments = **€105,000** (given directly).
Gain on disposal recognised = total potential gain × proportion of rights *transferred* = (fair value − carrying amount) × (1 − 21%) = (500,000 − 400,000) × 79% = **€79,000**.

Consistency check: asset-side changes = −400,000 (derecognise building) + 84,000 (recognise ROU asset) + 500,000 (cash received) = +€184,000. Liability-side + equity-side changes = +105,000 (lease liability) + 79,000 (gain, through profit or loss) = +€184,000. The two sides match exactly, confirming the calculation is internally consistent with how the transaction actually balances.

5. Deferred Taxes and Tax Rate Changes: The Adjustment Rule

Base assumption: deferred tax assets and liabilities must be based on *expected future* tax rates; absent other information, the current tax rate is generally assumed to continue into the future. If the government actually changes the statutory tax rate, existing DTA/DTL balances must be remeasured to reflect the new rate.

Tax rate change adjustment = Beginning balance of accumulated timing differences × (Former tax rate − New tax rate)

where: - the "beginning balance of accumulated timing differences" is the cumulative (not just current-year) temporary difference outstanding at the point the rate change takes effect; - the adjustment is applied once, as a catch-up, to the entire existing balance — not spread prospectively over future years.

Must know (direction of the adjustment, all flowing through Income Tax Expense):
• **Tax rate increases, DTA balance:** DTA **increases** — Dr DTA, Cr Tax Expense (a gain, tax expense falls)
• **Tax rate increases, DTL balance:** DTL **increases** — Dr Tax Expense, Cr DTL (a loss, tax expense rises)
• **Tax rate decreases, DTA balance:** DTA **decreases** — Dr Tax Expense, Cr DTA (a loss, typically effected via a valuation allowance)
• **Tax rate decreases, DTL balance:** DTL **decreases** — Dr DTL, Cr Tax Expense (a gain, tax expense falls)

- The intuition: a DTA is a future tax *saving*, so a higher future tax rate makes that saving more valuable (DTA increases); a DTL is a future tax *cost*, so a higher future rate makes that cost worse (DTL increases).
- Both move in the same direction as the rate change; a rate decrease moves both in the opposite direction.

6. Worked Example: A Full Tax-Rate-Change Life Cycle

Setup (adapted from the "Mike Mayo" exercise template): manufacturing equipment costing $600, 6-year life, no salvage value. Income before taxes and depreciation is $1,000 every year. Financial-reporting (book) depreciation is straight-line: $100/year for 6 years. Tax depreciation follows MACRS (200% declining balance with half-year convention): Year 1 = 120.00, Year 2 = 192.00, Year 3 = 115.20, Year 4 = 69.12, Year 5 = 69.12, Year 6 = 34.56 (summing to 600.00, fully depreciating the asset by end of Year 6, same as book depreciation's cumulative total). The tax rate is 40% in Years 1–3, then decreases to 35% for Years 4–6.

Watch out: the source material provides this as a blank journal-entry template (accompanying a separate Harvard case, "Mike Mayo Takes on Citigroup") without filled-in numbers. The full computation below is self-derived from the given depreciation schedule and the tax-rate-change formula in Section 5; every year's numbers were cross-checked against the cumulative depreciation-difference column for internal consistency (they reconcile exactly), but this is not a transcription of an official solution.

Years 1–3 (tax rate 40% throughout): book depreciation is 100 every year, so accounting profit = 1,000 − 100 = 900, and tax expense @40% = 360 every year. Tax depreciation exceeds book depreciation in each of these years (120, then 192, then 115.20 vs. book's flat 100), so taxable profit is progressively *lower* than accounting profit and a deferred tax **liability** builds: Year 1 taxable profit = 1,000 − 120 = 880, taxes payable = 352, deferred tax expense = 360 − 352 = 8 (DTL +8). Year 2 taxable profit = 1,000 − 192 = 808, taxes payable = 323.20, deferred tax expense = 36.80 (DTL +36.80, cumulative DTL = 44.80). Year 3 taxable profit = 1,000 − 115.20 = 884.80, taxes payable = 353.92, deferred tax expense = 6.08 (DTL +6.08, cumulative DTL = 50.88). By the end of Year 3, cumulative accumulated timing difference = 120+92+15.2 = 127.20, and cumulative DTL = 127.20 × 40% = €50.88, matching exactly.

Tax rate change at the start of Year 4 (40% → 35%): adjustment = beginning-balance accumulated timing difference (127.20) × (40% − 35%) = 127.20 × 5% = €6.36. Since the rate *decreased* and this is a DTL, the DTL **decreases**: Dr DTL 6.36, Cr Tax Expense 6.36 (a gain). New DTL balance = 50.88 − 6.36 = €44.52, which equals the same cumulative timing difference (127.20) now revalued at the new 35% rate (127.20 × 35% = 44.52) — exactly what the rate-change adjustment is designed to achieve.

Years 4–6 (tax rate 35%, temporary differences now reversing): from Year 4 onward, tax depreciation (69.12, 69.12, 34.56) falls *below* book depreciation (100 each year), so the accumulated timing difference starts unwinding and the DTL starts reversing. Year 4: accounting profit 900, tax expense @35% = 315; taxable profit = 1,000 − 69.12 = 930.88, taxes payable @35% = 325.81; deferred tax expense = 315 − 325.81 = −10.81 (DTL decreases by 10.81, balance 44.52 − 10.81 = 33.71). Year 5: identical to Year 4 (same 69.12 tax depreciation), DTL decreases by another 10.81, balance = 22.90. Year 6: tax depreciation 34.56, taxable profit = 1,000 − 34.56 = 965.44, taxes payable @35% = 337.90, deferred tax expense = 315 − 337.90 = −22.90 (DTL decreases by 22.90, balance = 22.90 − 22.90 = **0**). By the end of Year 6, both cumulative tax depreciation and cumulative book depreciation equal 600 (the full asset cost), the temporary difference has fully reversed, and the DTL is exactly zero — confirming, as in Session 20, that the deferred tax balance always fully unwinds by the end of the asset's life.

7. Common Mistakes and Exam Traps

- Recognising the full gain/loss on a sale-and-leaseback when the seller-lessee retains a right of use over part of the asset — only the portion of profit/loss relating to the rights actually *transferred* to the buyer-lessor is recognised; the rest is embedded in the ROU asset's initial measurement instead.
- Measuring the ROU asset at the fair value of the retained right of use, rather than at the corresponding *proportion of the previous carrying amount* — the ROU asset uses the seller-lessee's own pre-transaction book value, scaled down, not a fresh fair-value estimate.
- Treating every "sale and leaseback" as automatically qualifying for sale accounting — the entire treatment (Sections 2-3) branches on whether IFRS15's control-transfer test is actually met; if not, both parties instead account for the cash exchanged as a loan under IFRS9.
- Spreading a tax-rate-change adjustment prospectively over future periods — it is a one-time catch-up on the *existing* balance, recognised immediately in the period the rate change is enacted (or substantively enacted), not amortised forward.
- Getting the direction of the DTA/DTL rate-change adjustment backwards — remember DTAs and DTLs move in the *same* direction as the rate itself (both increase when the rate rises, both decrease when it falls), since a DTA is a future benefit (worth more at a higher rate) and a DTL is a future cost (worth more, i.e. worse, at a higher rate).

8. Must Know Absolutely

Must know:
• Sale and leaseback treatment hinges entirely on whether the transfer meets IFRS15's sale/control-transfer test
• If a sale: ROU asset = previous carrying amount × (PV of leaseback payments ÷ fair value); gain recognised = (fair value − carrying amount) × (1 − that same ratio); lease liability = PV of leaseback payments
• If not a sale: seller-lessee keeps the asset on its books, and the cash received is a loan (IFRS9 financial liability for the seller-lessee, financial asset for the buyer-lessor)
• Tax rate change adjustment = beginning cumulative timing difference × (old rate − new rate), recognised immediately through tax expense
• DTAs and DTLs move in the **same** direction as a tax rate change: both increase when the rate rises, both decrease when it falls
• Whatever a rate-change adjustment sets the DTA/DTL balance to must equal the cumulative timing difference revalued at the new rate — a useful self-check on any tax-rate-change calculation

Watch Out

Watch out: both worked numerical examples in this session (the building sale-and-leaseback, and the Mike Mayo tax-rate-change life cycle) were self-derived from blank templates/unsolved exercises in the source material. Both pass strong internal consistency checks (the sale-and-leaseback balances exactly; the tax-rate-change DTL unwinds to precisely zero by Year 6), but treat them as worked-through applications of the taught method rather than as verified official solutions — confirm against the instructor's own answers before the exam.